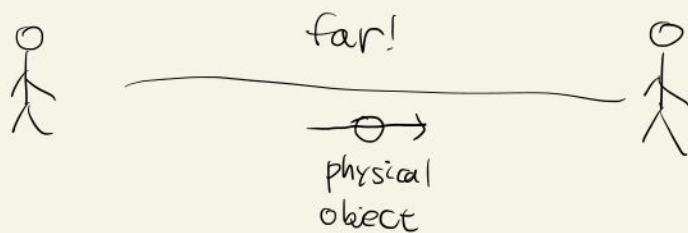


Ch7, Channel Capacity

how the C.C works?

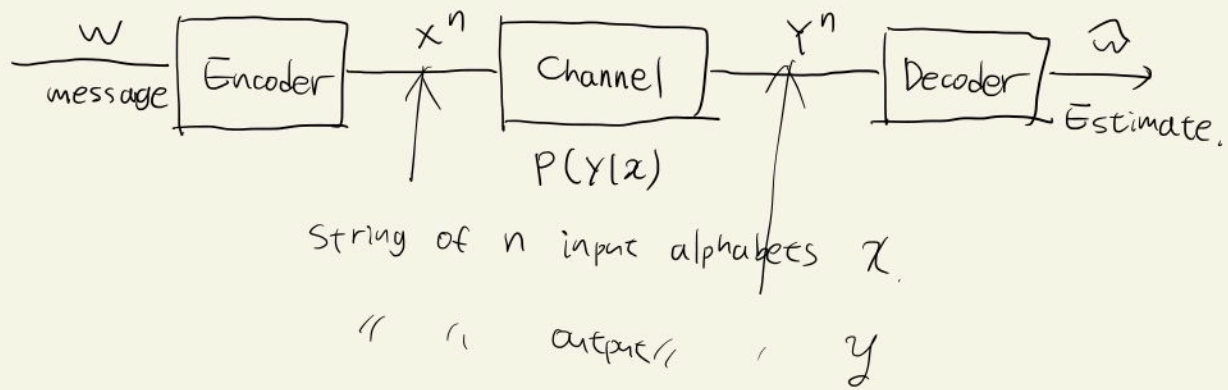
what is the channel capacity?



decide which alphabet
corresponds to each state.

noise, imperfections

Agree message $\rightarrow$ successful,



How many input signals are distinguishable for n uses of C.C? $\sim 2^{dn} \rightarrow$ channel capacity.

Shannon's second thm (C. coding thm)

$$\therefore d \Rightarrow C = \max_{P(x)} I(x, Y)$$

discrete memoryless channel (DMC)

• Noiseless Binary Channel

$$\begin{array}{ccc} 0 & \longrightarrow & 0 \\ x & & Y \\ 1 & \longrightarrow & 1 \end{array} \Rightarrow \text{Every signal of Alice} \rightarrow 2^n \cdot 1 \text{ distinguishable signals for } \underline{n} \text{ uses of } \underline{\text{C.C.}}$$

$$C = \max_{P(x)} I(x, Y) = \max_{P(x)} H(Y) - H(Y|x)$$

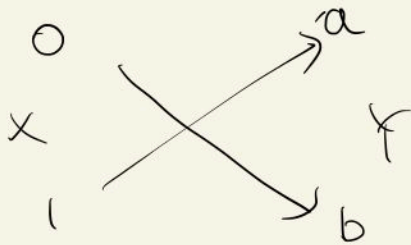
$H(Y|x)$: expected uncertainty of Y when x is given

$$\rightarrow H(Y|x) = 0$$

$$C = \max_{P(x)} H(x) = 1$$

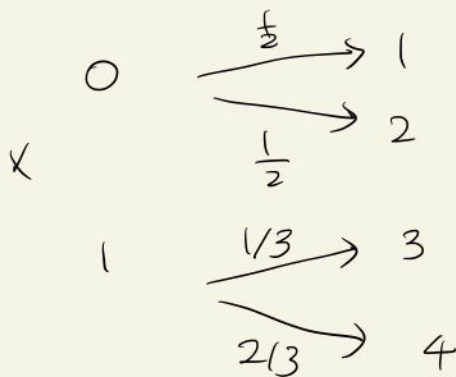
when x has uniform dist.

$$H(p) = -p \log_2 p - (1-p) \log_2 (1-p) = 1 \quad \text{when } p = \frac{1}{2}$$



every signal are distinguishable

- Noisy Channel with Nonoverlapping outputs.



Bob can always recover the exact input signal

$\Rightarrow$

133 | 243 | 214

$\Downarrow$
01100110001

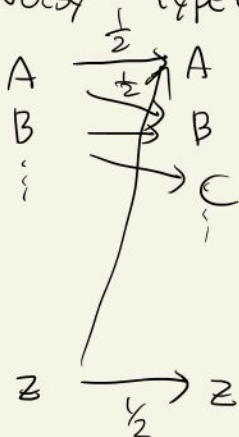
$\Rightarrow 2^n$ distinguishable, $d = 1$

$$\max_{P(x)} I(x, y) = \max_{P(x)} H(x) - \underbrace{H(x|y)} = \max_{P(x)} H(x) = 1$$

when $P(x) = (\frac{1}{2}, \frac{1}{2})$

* x is uniquely determined for given y

- Noisy Typewriter : input is either unchanged or changed to next letter with prob. 1/2.



$\Rightarrow$ If Alice only use

$A, C, E, \dots, Y \Rightarrow$ all signals of these kinds are distinguishable

But the possible output for this is 26^n .

$\Rightarrow 13^n$ distinguishable input signal

$$= 2^{n \log_2 13} \quad \text{channel capacity} = \log_2 13$$

$$\max_{p(x)} I(x, Y) = \max_{p(x)} (H(Y) - \underbrace{H(Y|X)})$$

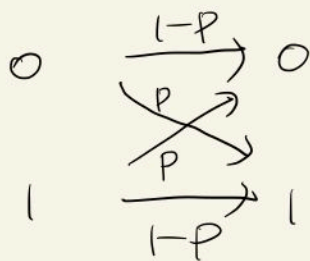
$$* \quad H(Y|X=x) = H\left(\frac{1}{2}\right) = 1 \Rightarrow H(Y|X) = 1$$

$$C = \max_{p(x)} \frac{H(Y)}{1} - 1 = \log_2 26 - 1 = \log_2 13$$

achieved when Y is uniform.

we can choose $p(x) = (\alpha, \frac{1}{13} - \alpha, \alpha, \frac{1}{13} - \alpha, \dots)$
for every $\alpha \in [0, \frac{1}{13}]$.

• Binary Symmetric Channel (BSC)



$$\begin{aligned} C &= \max_{p(x)} I(X, Y) \\ &= \max_{p(x)} (H(Y) - \underbrace{H(Y|X)}) \\ &= \max_{p(x)} H(Y) - H(p) \\ &= 1 - H(p) \end{aligned}$$

achieved when $p(x) = (\frac{1}{2}, \frac{1}{2})$

if $p = 0$ or $1 \rightarrow$ Noiseless Binary Channel

$$C = 1.$$

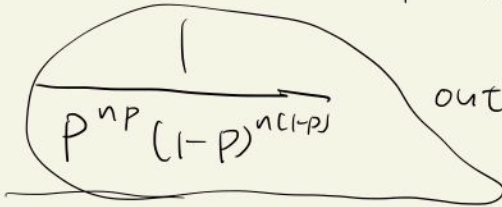
if $p = \frac{1}{2} \rightarrow$ Bob has no idea what the input was.

$$\underline{C} = 1 - H\left(\frac{1}{2}\right) = 0.$$

naive explanation;

For each transmission, the probability of error is p

If Alice send n letters, the Bob expects np wrong words and $n(1-p)$ correct words, each of them has probability $\underline{p^{np} (1-p)^{n(1-p)}}$

$\Rightarrow$ We approximate this to Noisy typewriter with 2^n alphabets,  outputs with uniform distribution.

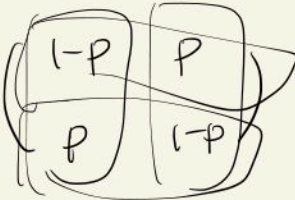
$\Rightarrow$ We expect $\frac{2^n}{1/p^{np} (1-p)^{n(1-p)}}$ distinguishable signals

$$= \left(2^{\frac{1 + p \log p + (1-p) \log (1-p)}{1 - H(p)}} \right)^n$$

• Symmetric Channels

A channel is symmetric if all rows & columns are permutations of each other

// weakly symmetric if all rows are perm, and all $\sum_x P(y|x)$ are equal,

BSC: symmetric  $\rightarrow$ symmetric

If C.C is weakly symmetric,

$$I(X, Y) = H(Y) - \underline{H(X|Y)} = H(X) - \underline{H(Y)}$$

$X|Y=y$: same for each y

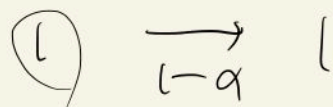
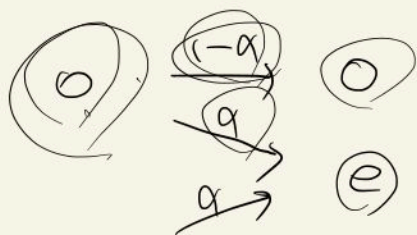
$$C = \max_{p(x)} H(X) - H(Y) = \log_2 |Y| - H(Y)$$

$\uparrow$
when Y is uniform

If we choose $p(x) = (\frac{1}{|X|}, \dots, \frac{1}{|X|}) \Rightarrow$

Y is uniform.

• Binary Erasure Channel



$$\begin{aligned} \Rightarrow C &= \max_{p(x)} I(X, Y) \\ &= \max_{p(x)} H(X) - \underline{H(X|Y)} \\ &= \max_{p(x)} \underline{H(X)} - H(Y) \end{aligned}$$

In fact, if $P(X=1) = p$, then

$$P(Y) = (\underline{(1-\alpha)(1-p)}, \alpha, (1-\alpha)p)$$

$$\rightarrow H(Y) = -(1-\alpha)(1-p) \log_2 (1-\alpha)(1-p) - \alpha \log_2 \alpha - (1-\alpha)p \log_2 (1-\alpha)p$$

$$= H(\alpha) + (1-\alpha) H(P)$$

$$\Rightarrow I(X, Y) = (1-\alpha) H(P) \quad : \text{ attains a max at } P = \frac{1}{2}$$

$$\Rightarrow C = 1-\alpha.$$

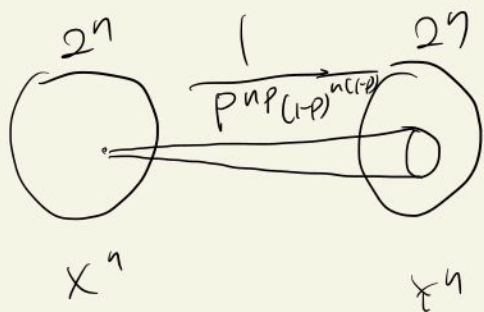
Channel Capacity; Bob is expected to receive $n(1-\alpha)$ words and $n\alpha$ erased words,

$\rightarrow 2^{n(1-\alpha)}$ words, we can relate one input signal, which are distinguishable

$$\rightarrow \text{Channel capacity} = 1-\alpha$$

These kinds of coincidence happen in general

BSC: For n use of channel, 2^n input signals, output signals,



$\approx$ Noisy typewriter with 2^n alphabets, each input has $p^n P(1-P)^{n(1-P)}$ outputs

$$\rightarrow \text{at most } 2^n p^n P(1-P)^{n(1-P)} \text{ distin. inputs} \\ = 2^{(1-H(P))n}$$

In general, for given C.C, we approximate

its n-blocks signal as a noisy typewriter (by using AEP)

$\Rightarrow$ each (typical) input has $\approx 2^{nH(X)}$ possible outputs
 $\left(\frac{1}{P^n P_{(1-P)}^{n(1-P)}} \right)$ in BSC)

Total possible (typical) output n-seq: $2^{nH(X)}$

$\Rightarrow$ We expect (at most) $2^{nH(X)} / 2^{nH(X)} = 2^{nI(X,Y)}$
 distinguishable input n-sequences

$$\Rightarrow C = \max_{P(X)} I(X,Y) \Rightarrow \downarrow$$

Roughly speaking, Bob decode Y^n as W if

$X^n(W)$ is "jointly typical" with Y^n .

The set $A_\epsilon^{(n)}$ (jointly typical pairs) w.r.t. $P(X,Y)$ is

$$A_\epsilon^{(n)} = \left\{ (x^n, y^n) \in \mathcal{X}^n \times \mathcal{Y}^n : \begin{aligned} & \left| -\frac{1}{n} \log P(x^n) - H(X) \right| < \epsilon \\ & \left| -\frac{1}{n} \log P(y^n) - H(Y) \right| < \epsilon \\ & \left| -\frac{1}{n} \log P(x^n, y^n) - H(X,Y) \right| < \epsilon \end{aligned} \right\}$$

$$P(x^n, y^n) = \prod_{i=1}^n P(x_i, y_i)$$

$$x^n = x_n x_{n-1} \dots x_1$$

$$y^n = y_n \dots y_1$$

$\Rightarrow$ We have

$$\textcircled{1} \quad \text{Prob}((x^n, y^n) \in A_\epsilon^{(n)}) \rightarrow 1 \quad \text{as } n \rightarrow \infty.$$

$$\textcircled{2} \quad |A_\epsilon^{(n)}| \leq 2^{n(I(x,y) + \epsilon)}$$

$$\textcircled{3} \quad \text{If } (x^n, y^n) \sim P(x^n) P(y^n),$$

$$\Rightarrow \text{Prob}((x^n, y^n) \in A_\epsilon^{(n)}) \leq 2^{-n(I(x,y) - 3\epsilon)}$$

and

if $n \gg 1$,

$$\text{Prob}((x^n, y^n) \in A_\epsilon^{(n)}) \geq (1-\epsilon) 2^{-n(I(x,y) + 3\epsilon)}$$

PF) $\textcircled{1}$ use LLN : we have

$$- \frac{1}{n} \log P(x^n) \rightarrow H(X) \quad \text{in probability}$$

$$- \frac{1}{n} \log P(y^n) \rightarrow H(Y) \quad \text{in } //$$

$$- \frac{1}{n} \log P(x^n, y^n) \rightarrow H(X, Y) \quad //$$

$\Rightarrow \forall \delta > 0, \exists n_0 \in \mathbb{N}$, s.t. $n > n_0$, then

$$\text{Prob} \left(\left| -\frac{1}{n} \log P(x^n) - H(X) \right| \geq \epsilon \right) < \delta.$$

$$\text{Prob} \left(\left| -\frac{1}{n} \log P(y^n) - H(Y) \right| \geq \epsilon \right) < \delta$$

$$\text{Prob} \left(\left| -\frac{1}{n} \log P(x^n, y^n) - H(X, Y) \right| \geq \epsilon \right) < \delta$$

$$\Rightarrow \text{Prob}((x^n, y^n) \notin A_\epsilon^{(n)}) < 3\delta$$

$$\Rightarrow \lim_{n \rightarrow \infty} \text{Prob}((x^n, y^n) \in A_\epsilon^{(n)}) = 1$$

$$\textcircled{2} \quad 1 \geq \sum_{(x^n, y^n) \in A_\epsilon^{(n)}} \frac{P(x^n, y^n)}{2^{-n(H(x, y) + \epsilon)}} \geq |A_\epsilon^{(n)}| 2^{-n(H(x, y) + \epsilon)}$$

$$\Rightarrow |A_\epsilon^{(n)}| \leq 2^{n(H(x, y) + \epsilon)}$$

$$\begin{aligned} \textcircled{3} \quad \text{Prob}((\tilde{x}^n, \tilde{y}^n) \in A_\epsilon^{(n)}) &= \sum_{(x^n, y^n) \in A_\epsilon^{(n)}} P(x^n) P(y^n) \\ &\leq 2^{n(H(x, y) + \epsilon)} 2^{-n(H(x) - \epsilon)} 2^{-n(H(y) - \epsilon)} \\ &= 2^{-n(I(x, y) - 3\epsilon)} \end{aligned}$$

lower bound: for large n ,

$$\underline{1 - \epsilon} \leq \text{Prob}(A_\epsilon^{(n)}) = \sum_{(x^n, y^n) \in A_\epsilon^{(n)}} P(x^n, y^n)$$

$$\leq |A_\epsilon^{(n)}| 2^{-n(H(x, y) - \epsilon)}$$

$$\Rightarrow |A_\epsilon^{(n)}| \geq (1 - \epsilon) 2^{n(H(x, y) - \epsilon)}$$

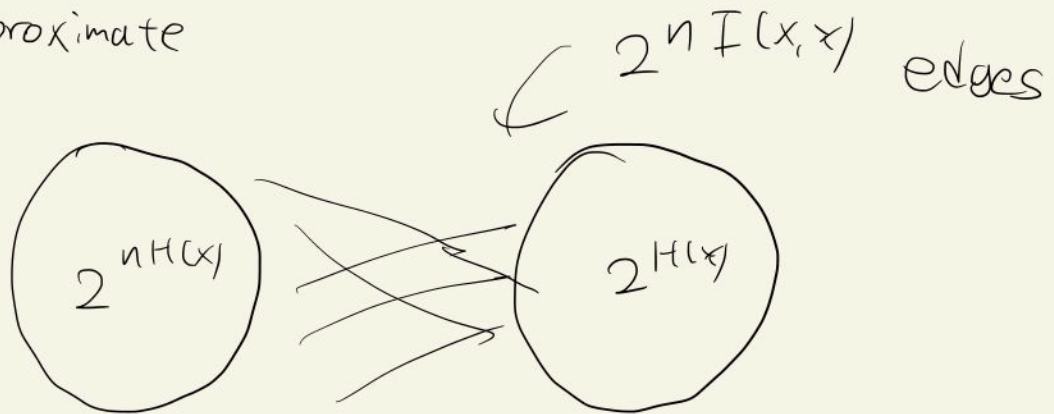
$$\Rightarrow \text{Prob}((X^n, Y^n) \in A_\epsilon^{(n)}) = \sum_{(x^n, y^n) \in A_\epsilon^{(n)}} P(x^n) P(y^n)$$

$$\geq (1-\epsilon) 2^{n(H(X|Y)-\epsilon)} \times 2^{-n(H(X)+\epsilon)} 2^{-n(H(X)+\epsilon)}$$

$$= (1-\epsilon) 2^{-n(I(X,X)+3\epsilon)}$$

* Prob. of jointly typical pair $\approx 2^{-nI(X,X)}$

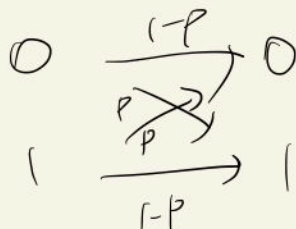
We approximate



$\Rightarrow$ We expect at most $2^{nI(X,Y)}$ distinguishable input signals, $C.C = \max_{P(a)} I(X,X)$

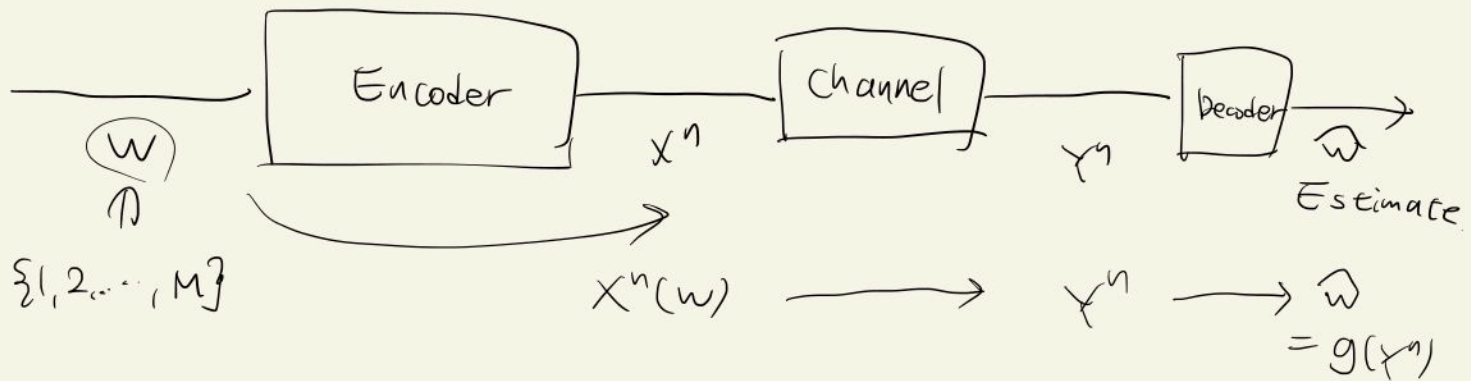
Since these argument uses a naive AEP Approx, there might be a channel s.t. $P(y|x) > 0 \forall x,y$

BSC



We require Bob decode output with arbitrarily small error
(instead of distinguish)

Recall that Alice send a message w ,



Bob makes an error if $\hat{w} = g(y^n) \neq w$

• A discrete channel, denoted by $(\mathcal{X}, P(y|x), \mathcal{Y})$,

$$\text{s.t. } P(y|x) \geq 0, \sum_y P(y|x) = 1 \quad \forall x \in \mathcal{X}$$

• n-th extension of discrete memoryless channel (DMC)

$$\text{is } (\mathcal{X}^n, P(y^n|x^n), \mathcal{Y}^n) \quad \left(P(y_k|x^n, y^{k-1}) = P(y_k|x_k) \right)$$

* If Bob does not give a feedback to Alice,

$$P(x_k|x^{k-1}, y^{k-1}) = P(x_k|x^{k-1}), \text{ then}$$

$$P(y^n|x^n) = \frac{P(y_1, y_2, \dots, y_n, x_1, x_2, \dots, x_n)}{P(x_1, x_2, \dots, x_n)} = \frac{P(y_1|x_1)P(y_2|x_2, y_1) \cdots P(y_n|x_n, y^{n-1})}{P(x_1)P(x_2|x_1) \cdots P(x_n|x^{n-1})}$$

$$= P(y_1|x_1) P(y_2|x_2, y_1) \cdots P(y_n|x_n, y^{n-1})$$

$$= P(y_1|x_1) P(y_2|x_2) \cdots P(y_n|x_n)$$

• An (M, n) code for $(\mathcal{X}, P(\mathcal{X}), \mathcal{Y})$ consists of

① Index set $\{1, 2, \dots, M\}$

② Encoding fcn $x^n: \{1, 2, \dots, M\} \rightarrow \mathcal{X}^n$
 $i \mapsto x^n(i)$

range = codebook

③ Decoding fcn $g: \mathcal{Y}^n \rightarrow \{1, 2, \dots, M\}$ deterministic

• λ_i : conditional prob. of error given that i was chosen by Alice.

$$\lambda_i = \text{Prob}(\underbrace{g(y^n)}_{\neq i} \mid x^n = \underbrace{x^n(i)})$$

$$\lambda^{(n)} := \max_{1 \leq i \leq M} \lambda_i$$

• $P_e^{(n)}$: average of error for (M, n) code

$$= \frac{1}{M} \sum_{i=1}^M \lambda_i$$

• The rate R of (M, n) code

$$:= \frac{\log_2 M}{n} \quad (\text{bits per transmission})$$

• The rate R is said to be achievable if

if there is a seq. of $(\lceil 2^{nR} \rceil, n)$ codes, s.t.

$$\lim_{n \rightarrow \infty} \lambda^{(n)} = 0$$

relaxed version of distinguishable signals

• The capacity of a channel is the supremum

of all achievable rates,

Thm) (Shannon's Channel coding thm)

For a DMC, all rates below $C = \max_{p(x)} I(x, y)$ are achievable.

Specifically, for every $R < C$, there exists a seq. of $(\lceil 2^{nR} \rceil, n)$ codes with $\lim_{n \rightarrow \infty} \lambda^{(n)} = 0$

Conversely, any sequence of $(\lceil 2^{nR} \rceil, n)$ codes with $\lim_{n \rightarrow \infty} \lambda^{(n)} = 0$ must have $R \leq C = \max_{p(x)} I(x, y)$